AI Marketing Operating System

Final Product Research, Architecture, UI/UX, and Development Plan

Prepared for	Sujit Waiba Lama
Document purpose	Record the final direction of the AI marketing system and convert research into a buildable development plan, including AI generation, user-uploaded media scheduling, and reference-design-guided creative generation.
Version	1.2 - Updated user media upload and reference design scope
Date	2 June 2026
Main decision	Build an image-first, human-approved, Meta-first creative operations platform with AI copilot, scheduling, performance learning, support for user-uploaded media, and reference design/style guidance.
Primary stack direction	Next.js + Supabase/PostgreSQL + OpenAI Image API + Meta Graph APIs, with video providers abstracted for later phases.

Core principle: AI drafts. Human approves. Data improves.

1. Executive Summary

This document finalizes the product direction for an AI-powered marketing operating system. The system should not be a generic AI image generator or a direct clone of Higgsfield. The recommended product is a focused creative operations platform where small businesses, education consultancies, local brands, and small agencies can plan campaigns, generate image-first marketing assets, manually refine them, approve them, schedule posts, and learn from performance data.

The strongest opportunity is the gap between creative generation and marketing execution. Higgsfield is strong at high-quality media generation. Tools such as Sprout Social, Hootsuite, Canva, and Adobe focus on workflow, scheduling, analytics, and brand control. The proposed system should sit in the middle: it should combine AI creative assistance with human review, brand governance, publishing calendars, and performance feedback.

Final decision	What it means for the build
Do not build a full Higgsfield clone	Competing on cinematic video generation, avatars, and massive model access is not realistic for one developer.
Start image-first	Static creatives are cheaper, easier to review, easier to edit, and easier to schedule than video.
Use an AI copilot, not chat-only UX	Chat should help users create and control campaigns, but the main workspace must show posts, assets, approvals, calendar, and analytics.
Keep human approval	Users should approve before content is posted. This improves trust and reduces brand/safety risk.
Go Meta-first	Facebook Page and Instagram Business workflows are the most practical starting point for social publishing and analytics.
Build the learning loop	The system should tag creative variants and connect them to performance so future recommendations improve.
Support user-uploaded media	Users can upload their own images or videos, add captions, approve them, and schedule them for publishing. AI generation is optional, not mandatory.
Support reference designs	Users can upload preferred/reference designs so the AI can understand desired style, layout, tone, and visual direction while still producing original outputs.

2. How the Idea Was Formed

The idea developed from observing a real change in marketing work. Businesses increasingly need more content, more variations, faster creative testing, and consistent brand execution. At the same time, AI generation tools can create images and videos quickly, but most of them stop at generation. They do not fully solve the operational workflow: planning, editing, approval, scheduling, reporting, and learning from results.

- The first idea was broad: an AI system that could help run marketing work.
- Research showed that this was too large and risky for an early product.
- The idea was narrowed into an image-first creative operations system.
- Higgsfield research showed the value of product intake, creative presets, batch generation, and AI-agent workflows.
- Social media management research showed that teams still pay for calendars, approvals, reporting, and brand control.
- The final direction became a practical platform: campaign planning + AI generation + manual editing + approval + scheduling + analytics.

This origin matters because it prevents the product from becoming another prompt box. The product should guide users through marketing decisions and convert those decisions into real posts, assets, and schedules.

3. Research Foundation and Key Findings

The research uses the uploaded AI Marketing Operating System Research PDF, the latest Higgsfield overview notes, and public sources from Higgsfield, OpenAI, Meta, and UX research. The important conclusion is that creative generation alone is not enough. The useful product is a workflow system that makes creative output repeatable, governable, and measurable.

Research source	Key finding	Impact on product decision
Uploaded research PDF	A direct clone of AI ad generation is risky; the stronger opportunity is workflow and feedback loop.	Position the product as creative operations, not only generation.
Higgsfield Marketing Studio	Product URL/input + creative format + avatar/mode generates publish-ready ads.	Add product/service intake and creative presets.
Higgsfield Supercomputer	Agentic workflows can run scheduled creative tasks and use reusable skills.	Add an AI copilot and later add repeatable campaign actions.
OpenAI Higgsfield case study	Planning-first generation reduces prompt iteration and maps user intent into narrative/creative structure.	Build an AI planner before image generation.
Meta developer docs	Facebook/Instagram publishing and ads analytics are possible but require platform rules and permissions.	Start with Meta-first integration and keep scope narrow.

NN/g AI UX guidance	Prompt controls and structured UI help users discover what AI can do.	Use buttons, cards, templates, and guided choices around chat.

4. Competitor and Inspiration Analysis

4.1 What to Learn from Higgsfield

Higgsfield is valuable to study because it shows how AI creative products are becoming workflow systems. Their Marketing Studio reduces prompt effort by using product input, preset modes, avatars, hooks, and batch creative generation. Their Supercomputer points toward a future where users brief an agent and the agent orchestrates tools and workflows.

Higgsfield pattern	Why it works	How to adapt safely
Product/service intake	The user does not start from an empty prompt. The product context is structured first.	Allow manual product/service profile first; add URL import later.
Creative modes/presets	Modes such as UGC, tutorial, unboxing, TV spot reduce decision fatigue.	Use image-first presets: product launch, offer poster, testimonial, before/after, festival campaign.
Batch variations	Marketers need many variants for testing.	Generate 3-5 image variants per post in MVP, not hundreds.
Brand-aware generation	Brands need consistent tone, logo, colors, and safe output.	Build a Brand Kit and require prompts to use brand rules automatically.
Agentic workflows	Complex work becomes easier when users can ask for a complete workflow.	Add AI Copilot commands such as Create 7-day campaign and Improve weak post.
Scheduled tasks	Automation can run repeated creative tasks.	In MVP, schedule posts; later schedule creative generation tasks.

4.2 What Not to Copy

- Do not copy Higgsfield branding, UI layout, wording, icons, product names, or proprietary prompts.
- Do not compete directly on cinematic video generation in version 1.
- Do not promise competitor performance data that official APIs do not expose.
- Do not make Higgsfield your core backend unless API terms, pricing, rate limits, and commercial access are verified.
- Do not build too many modules at launch. A focused MVP beats an unfinished super-app.

5. Final Product Vision

The final product should be an AI marketing command center for small businesses and small teams. It should help users turn a business goal into ready-to-review social media creatives and then schedule those creatives for publishing. The product should feel simple for non-technical users, but internally it should have a structured workflow and data model.

Element	Final decision
Product promise	Create, upload, guide with references, approve, schedule, and improve social media creatives from one workspace.
Primary users	Small businesses, education consultancies, local brands, solo marketers, and small agencies.
First content type	Static AI-generated image posts plus user-uploaded images/videos for scheduling and publishing.
First platforms	Facebook Page and Instagram Business/Creator accounts.
AI role	Copilot and planner, not uncontrolled autonomous publisher.
Human role	Final reviewer and approver before scheduling or publishing.
Long-term moat	Brand memory + approval history + creative performance learning loop.
Content source rule	Every post can come from AI generation, user upload, or manual text/caption input, then follow the same approval and scheduling flow.
Reference design rule	Reference designs are used as inspiration and style guidance only. The system should extract style signals instead of directly copying another design.

6. Problem Statement

Small businesses and local teams struggle to maintain consistent social media marketing because the work is spread across many tools. They plan in notes or spreadsheets, design in Canva or similar tools, write captions separately, send assets for approval in chat apps, schedule posts in another tool, and then check performance manually. AI image generation helps with creation, but it does not automatically solve the full workflow.

- Content planning is inconsistent and often last-minute.
- Brand assets such as logo, colors, and tone are not enforced automatically.
- AI outputs often need manual correction, especially text, phone numbers, location, and CTA details.

- Approval happens informally through chat, causing confusion and rework.
- Scheduling is separate from creative generation.
- Performance data is rarely connected back to creative decisions.
- Users do not always know how to write strong prompts or identify which creative angle is best.

7. Proposed Solution

The proposed solution is a visual marketing workspace with an AI copilot. The user can create a brand workspace, add product/service details, choose a campaign goal, let AI suggest angles and post ideas, generate image variants, manually edit the final creative, approve it, schedule it, publish it, and learn from analytics.

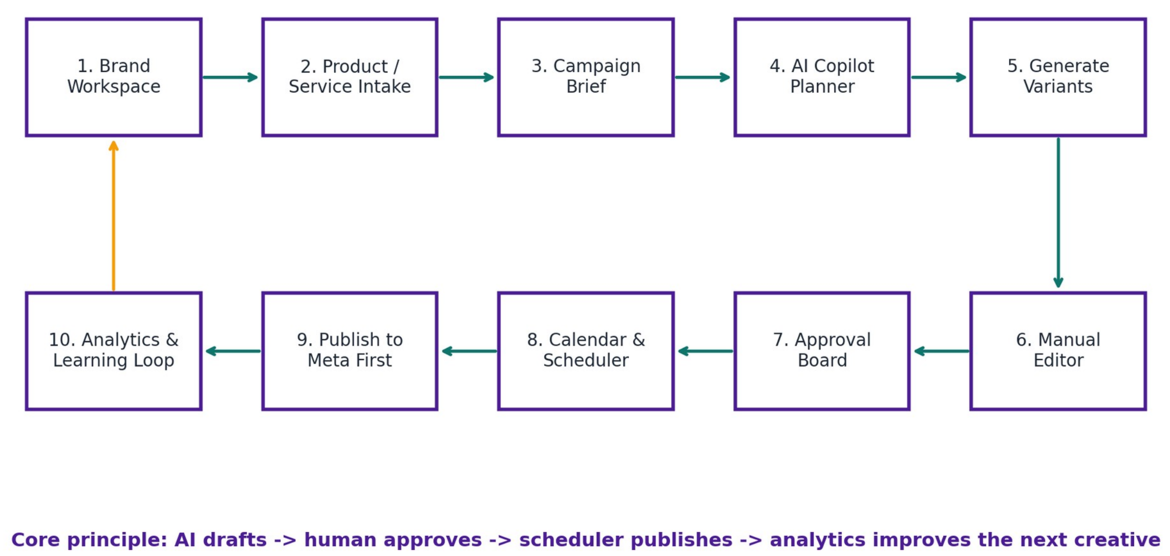

Figure 1. Recommended end-to-end product workflow.

8. UI/UX Direction

The best UX is not chat-only. The recommended structure is a dashboard and visual workflow with an AI Copilot side panel. Chat should help the user express goals, but the system should convert the conversation into concrete objects such as campaigns, posts, captions, image variants, approval items, and scheduled posts.

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Recommended layout

Left sidebar: Navigation
Center: Visual workspace
Right side: AI Copilot

Main pages:
Dashboard | Campaigns | Calendar | Brand Kit | Products/Services | Assets | Approvals |
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Analytics | Settings
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UX area	What the user sees	Why it matters
Dashboard	Today posts, pending approvals, AI recommendations, recent campaigns, top-performing content.	Gives instant clarity and reduces confusion.
AI Copilot	Chat with quick action buttons: Create campaign, Generate captions, Improve design, Fill calendar, Analyze performance.	Makes AI easier to use without forcing perfect prompts.
Campaign Builder	Step-by-step guided form: goal, audience, product, platform, style, schedule.	Turns vague marketing ideas into structured input.
Image Workspace	Generated variants, captions, hooks, status, edit/regenerate/approve buttons.	Keeps creative control visual.
Manual Editor	Text overlay, logo, colors, CTA, phone number, image crop, export size.	Protects quality and allows local business details.
Approval Board	Draft, Review, Needs Changes, Approved, Scheduled, Published.	Creates a real workflow, not only generated output.
Calendar	Month/week view with post thumbnails, platform icons, status, and time.	Makes scheduling clear and manageable.
Analytics	Best post, weak post, best hook, best time, recommendations.	Closes the loop and makes the product sticky.

9. Final Feature Scope

9.1 MVP Features

Feature module	MVP details
User authentication	Sign up, login, workspace selection, basic profile.
Brand Kit	Business name, logo, colors, tone of voice, target audience, language preferences, brand rules, and optional reference designs.
Product/Service Profile	Manual product/service details and image upload. URL import can be added after MVP.

Campaign Brief	Goal, audience, platform, offer, CTA, style, number of posts.
AI Campaign Planner	Generate angles, hooks, captions, and recommended post structure.
Image Generation	Generate 3-5 static image variants per post using brand, campaign, and optional reference design context. This is optional if the user uploads their own media.
Manual Editor	Edit text, color, logo placement, CTA, and crop/resize for platform sizes.
Approval Workflow	Draft, in review, needs changes, approved, scheduled, published, failed.
Calendar Scheduler	Schedule approved AI-generated posts or user-uploaded images/videos for Facebook/Instagram.
Publishing Integration	Meta-first publishing for Facebook Pages and Instagram Business/Creator accounts, supporting media posts where permissions allow.
Basic Analytics	Reach, impressions, engagement, clicks where available; store by post and variant.
Learning Notes	AI summarizes which hooks, formats, colors, or angles performed best.
User Media Upload	Upload user-produced images/videos, store them in the media library, add captions, approve, schedule, and track publish status.
Media Asset Library	Central place for generated images, uploaded images, uploaded videos, captions, thumbnails, source type, and platform-ready versions.
Reference Design Library	Upload preferred designs, brand examples, competitor inspiration, previous posters, or layout references. Store them as guidance assets for AI and manual design decisions.
Reference Style Extraction	AI summarizes reference design signals such as layout, color mood, typography direction, spacing, content hierarchy, CTA placement, and tone.

9.2 Features to Delay

- Full autonomous AI agent that publishes without review.
- AI avatar training and advanced UGC video generation.
- All-platform posting at launch.
- Competitor performance tracking beyond public creative observation.
- Enterprise SSO, complex team permission matrix, and advanced billing.
- Custom model training and private GPU infrastructure.

9.3 Content Source Scope: AI and User-Uploaded Media

The system should not force every post to be AI-generated. Many businesses already have their own photos, product videos, event clips, or edited creatives. Therefore, the platform must support both AI-created content and user-produced media. The same review, calendar, publishing, and analytics workflow should apply to every content source.

Content source	What the user provides	How the system handles it
AI-generated image	Campaign brief, brand kit, product details, style/template selection.	AI generates variants; user edits, approves, schedules, and publishes.
User-uploaded image	Existing photo, poster, product shot, or designed creative.	System stores it, creates thumbnail, lets user add caption, approve, schedule, and publish.
User-uploaded video	Existing short video, reel, event clip, product demo, or business-produced ad.	System stores it, validates platform requirements where possible, lets user add caption, approve, schedule, and publish.
Manual text/caption post	Text-only announcement, caption, offer, or link post.	System saves the post, places it in approval/calendar workflow, and publishes if the selected platform supports it.

Important implementation rule: user-uploaded video scheduling is different from AI video generation. AI video generation can be delayed to a later phase, but uploading an existing user video and scheduling it can be included earlier because the system is only storing, validating, and publishing the file.

Upload/ generate	Add caption	Approve	Schedule	Auto-publish	Track result
AI image or own media	Text, hashtags, CTA	Human review	Date, time, platform	Meta APIs / platform APIs	Published or failed + metrics

9.4 Reference Design and Style Guidance Scope

Users should be able to provide preferred reference designs because many non-technical users struggle to describe visual style using text alone. A reference design can be a previous brand poster, social media creative, screenshot, competitor inspiration, moodboard image, Canva/Figma export, product photo layout, or any visual example that communicates the user's desired direction. The system should use references as guidance, not as material to copy exactly.

Reference type	Example input	How the system should use it
Brand reference	Old posters, logo usage examples, brand campaign screenshots.	Extract brand-consistent layout, color usage, tone, and CTA style.
Design inspiration	A design the user likes from Pinterest, Instagram, Behance, Canva, or competitors.	Understand mood and structure, then generate an original version with different copy/layout details.
Template reference	User-uploaded layout example or preferred ad/post format.	Use as a structural guide for content hierarchy and placement.
Product photography reference	Preferred product angle, lighting, background, or scene.	Guide future image prompts and manual editing choices.
Campaign reference	A past campaign that performed well.	Connect style tags with performance learning for future recommendations.

The AI Copilot should ask simple questions after a reference is uploaded: "What do you like about this design?" and "Should I follow the colors, layout, mood, or typography?" This prevents the system from guessing incorrectly and makes the user feel in control.

Upload reference	Tag style	AI analyzes	User confirms	Generate variants	Manual edit	Save learning
Image/ screenshot/ design	Bold, clean, premium, local, etc.	Layout, colors, text hierarchy	Accept or adjust style notes	Original new creatives	Fix text/logo/CTA	Store as brand rule

Legal and ethical guardrail: the system should not directly clone another company's creative, logo, copyrighted layout, or campaign wording. It should convert references into high-level style attributes and generate an original creative for the user's own brand.

10. System Architecture

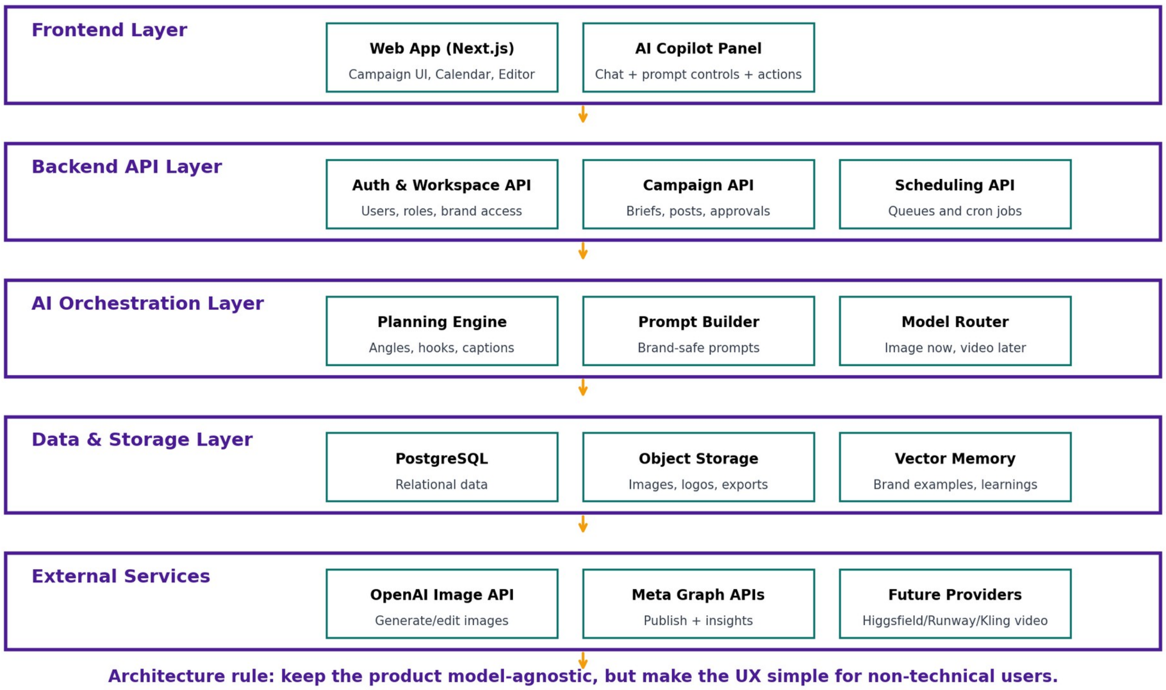

Figure 2. High-level system architecture for the proposed MVP and later scale-up.

10.1 Recommended Technology Stack

Layer	Recommended choice	Reason
Frontend	Next.js + React + Tailwind CSS	Fast to build, good UI control, easy deployment on Vercel.
Backend	Next.js API routes or Node.js/Express	Simple for one developer and enough for MVP.
Database	Supabase PostgreSQL	Relational data fits workspaces, campaigns, posts, approvals, and analytics.
Authentication	Supabase Auth or Clerk	Faster than building auth from scratch.
Storage	Supabase Storage or Cloudflare R2	Store logos, generated images, exports, thumbnails.
AI planning/text	OpenAI or similar LLM provider through an abstraction layer	Generates briefs, hooks, captions, and recommendations.
AI image	OpenAI Image API first, model-agnostic wrapper	Supports generation/editing and is easier than video for MVP.
Scheduling	Cron jobs / Supabase Edge	Background scheduling and

	Functions / Trigger.dev / Inngest	retry jobs.
Publishing	Meta Graph APIs first	Facebook and Instagram are practical first platforms for target users.
Analytics	Meta Insights / stored organic and paid metrics	Connect creative variants to performance.

11. Database and Data Model

Table	Purpose	Key fields
users	Stores user profile and login connection.	id, name, email, created_at
workspaces	Business or client workspace.	id, owner_id, name, business_type, timezone
workspace_members	Team access in later versions.	workspace_id, user_id, role
brands	Brand identity and rules.	workspace_id, logo_url, colors, tone, language, audience
brand_assets	Uploaded logos, fonts, examples, templates, and brand files.	workspace_id, asset_type, file_url, metadata
products	Product/service being promoted.	workspace_id, name, description, images, price, benefits
campaigns	Marketing campaign container.	workspace_id, product_id, goal, audience, platform, start/end dates
creative_templates	Reusable post/ad structures.	name, format, prompt_schema, layout_rules
posts	Individual post to generate/schedule.	campaign_id, platform, caption, media_asset_id, source_type, reference_asset_ids, status, scheduled_at
post_variants	Generated variants for one post.	post_id, hook_type, visual_style, cta_type, offer_type, generated_prompt, image_url, reference_asset_ids, is_selected
approvals	Review workflow history.	post_id, reviewer_id, status, comment, created_at
scheduled_jobs	Publishing queue.	post_id, media_asset_id, scheduled_at, platform, status, retry_count, error_message

published_posts	External platform IDs after publishing.	post_id, platform_post_id, published_at, permalink
analytics_metrics	Performance data over time.	post_id, impressions, reach, likes, comments, shares, clicks, engagement_rate
creative_learnings	AI-generated learning notes.	workspace_id, insight_type, summary, evidence, created_at
media_assets	Stores AI-generated and user-uploaded images/videos used in posts.	workspace_id, source_type, file_type, file_url, thumbnail_url, metadata, uploaded_by
reference_assets	Stores user-provided reference designs, screenshots, examples, and extracted style notes.	workspace_id, brand_id, file_url, label, tags, source, extracted_style_json, rights_status, created_by
style_rules	Stores confirmed design preferences derived from reference designs and user feedback.	workspace_id, brand_id, rule_type, rule_text, source_reference_id, confidence_score

12. AI Pipeline Design

The AI pipeline should be structured, not a single raw prompt. The system should turn user input into a campaign brief, combine it with brand rules, uploaded media, and optional reference design notes, then produce image prompts, captions, variants, and scheduled posts that can be reviewed and improved over time.

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AI pipeline

1. Collect structured input
   Brand + product + campaign goal + audience + platform + style + optional reference design

2. Decide content source
   AI-generated image, user-uploaded image/video, or manual text/caption post

3. Generate campaign strategy
   Angles, hooks, post ideas, captions, recommended schedule

4. Build image prompt or media draft
   Brand rules + template rules + product details + platform size + reference style notes

5. Generate or attach creative
   Generate 3-5 AI image variants or attach user-uploaded media to a post draft
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6. Score, edit, and approve
   Brand fit, clarity, hook strength, CTA clarity, platform fit, originality check

7. Publish and learn
   Metrics are stored and AI suggests the next better creative

AI component	Function	MVP approach
Campaign Planner	Turns campaign brief into angles, hooks, captions.	LLM prompt with structured JSON output.
Prompt Builder	Combines brand kit and template rules into image prompt.	Template-based prompt schemas.
Image Generator	Creates static creative variants.	OpenAI Image API first.
Creative Scorer	Scores clarity, brand fit, hook strength, CTA quality.	LLM rubric first; later combine with real performance data.
Learning Summarizer	Explains what worked and what to try next.	Analyze stored analytics + creative tags.
Media intake handler	Accepts user-uploaded images/videos, stores files, creates thumbnails, and attaches media to posts.	Use storage buckets, file validation, and source_type tracking; do not treat uploaded content as generated content.
Reference analyzer	Reads user-uploaded reference designs and extracts style signals like layout, mood, color usage, typography direction, CTA hierarchy, and image composition.	MVP can use a vision-capable LLM to summarize style into JSON notes; avoid exact copying.
Style memory	Stores user-confirmed design preferences from references and approvals.	Use style_rules and reference_assets to keep future generations aligned without directly copying the reference.

13. Final User Journey

Step	User action	System action	Output
1	Create workspace	Stores business profile and timezone.	Workspace dashboard.
2	Add brand kit	Stores logo, colors, tone, language, audience.	Brand rules used in AI prompts.
3	Set brand kit and references	Stores logo, colors, tone, language, brand rules, and optional preferred reference	Brand memory and style guidance.

		designs.	
4	Create campaign	Collects goal, audience, platform, style, number of posts.	Campaign brief.
5	Ask AI Copilot	Generates angles, hooks, captions, and post plan.	Campaign plan.
6	Generate or upload creative	Creates AI image variants using optional references or stores user-uploaded image/video as a media asset.	Visual variants or uploaded media draft.
7	Manual edit	User edits headline, CTA, logo, color, crop.	Final creative draft.
8	Approve	Moves post through review state.	Approved post.
9	Schedule	Creates publishing job with selected media, caption, platform, date, and time.	Calendar item with scheduled status.
10	Publish	Calls Meta APIs at scheduled time.	Published social post.
11	Analyze	Collects metrics and compares creative tags.	Learning notes and next recommendations.

14. API Integration Plan

Integration	Use in system	MVP notes
OpenAI Image API	Generate and edit image creatives.	Use direct API first; store prompts and outputs for traceability.
Meta Pages API	Publish and manage Facebook Page posts.	Requires Facebook app setup, permissions, page access tokens.
Instagram Content Publishing API	Publish images/videos/reels/carousels to Instagram Business/Creator accounts.	Requires connected IG Business/Creator account and Facebook Page.
Meta Ads Insights API	Get paid campaign performance data.	Useful after users connect ad accounts; not needed on day 1 if starting organic.
Storage/CDN	Serve generated images, user-uploaded images/videos,	Use signed URLs for private assets; store source_type and

	reference designs, thumbnails, and platform-ready media.	file metadata.
Email/notifications	Approval requests, failed post alerts, weekly performance summaries.	Can use Resend or similar later.
Media validation layer	Checks file type, size, duration, and platform-specific readiness before scheduling.	Keep simple in MVP: validate images/videos and show clear errors if a platform cannot publish the file.
Reference style analysis	Processes reference images/screenshots and stores extracted style notes for generation.	Use AI vision analysis and user confirmation; do not generate direct copies.

15. Development Breakdown

Break the development into small modules so the work does not feel overwhelming. Each module should be testable by itself.

Part	Module	Main tasks	Definition of done
Part 1	Project setup	Next.js app, Tailwind, Supabase project, environment variables, GitHub repo.	App runs locally and deploys to Vercel.
Part 2	Auth + workspace	Login, signup, workspace creation, basic settings.	User can create a business workspace.
Part 3	Brand Kit	Logo upload, colors, tone, audience, language preference, brand rules, reference design upload.	Brand data and reference designs are stored and visible.
Part 4	Product/Service profile	Manual product/service input, image uploads, benefits, price/offer fields.	Product can be reused in campaigns.
Part 5	Campaign builder	Accept campaign brief, brand kit, product details, optional reference style notes, and generate 3-5 image variants.	Campaign brief is created.
Part 6	AI Copilot planner	Generate hooks, captions, angles, post	AI returns structured

		plan from brief.	campaign plan.
Part 7	Image generator	Build prompt from brand + product + template; call image API; store variants.	User sees 3 image variants.
Part 8	Manual editor	Text/logo/color/CTA editing and export size.	User can make simple practical changes.
Part 9	Approval workflow	Status transitions, comments, approve/reject.	Approved posts are separated from drafts.
Part 10	Calendar scheduler	Month/week calendar, scheduled_at field, queue job, support uploaded or generated media.	Approved AI or uploaded post appears in calendar.
Part 11	Meta publishing	Connect page/account, publish scheduled post, store external IDs.	Test post publishes to Facebook/Instagram sandbox or test page.
Part 12	Analytics + learning	Fetch metrics, store results, AI summary of learnings.	Dashboard shows post performance and next recommendation.
Part 4A	Media upload library	Image/video upload, thumbnails, source_type, media metadata, link media to posts.	User can upload own image/video and reuse it in a scheduled post.
Part 4B	Reference design library	Reference upload, tags, style notes, AI analysis, link reference to brand/campaign/post.	User can upload a preferred design and use it to guide future creative generation.

16. Suggested 10-Week Development Timeline

Week	Goal	Output
1	Foundation	Repo, design system, auth, workspace shell.
2	Brand and product setup	Brand Kit and Product/Service screens completed.
3	Campaign builder	Guided campaign brief flow and database tables.
4	AI planner	AI generates campaign angles, captions, hooks, and post ideas.

5	Image generation	Generate and store image variants with prompt metadata.
6	Manual editor	Basic editing: headline, CTA, logo, colors, resize/export.
7	Approval board	Draft/review/approve statuses and comments.
8	Calendar scheduler	Month/week view and scheduled jobs.
9	Meta integration	Connect test page/account and publish approved scheduled posts.
10	Analytics and polish	Basic metrics, learning notes, dashboard polish, testing, documentation.

17. Page-by-Page UI Requirements

Page	Required components
Dashboard	Quick create button, pending approvals, scheduled posts, top campaign, AI recommendations.
AI Copilot	Chat panel, quick action buttons, contextual suggestions, campaign planning, caption help, reference-style questions, analytics explanation.
Brand Kit	Logo upload, color picker, tone selector, language selector, brand rules, reference design upload, style notes, and reference examples.
Products/Services	Product list, add product/service, upload product images/videos, benefits, target customer, offer.
Campaign Builder	Goal selector, audience selector, platform selector, style preset, post count, schedule preference.
Creative Workspace	Variant grid, selected preview, uploaded media preview, reference design panel, caption editor, regenerate button, manual edit button, approve button.
Manual Editor	Canvas preview, text controls, logo placement, colors, CTA, image crop/replace, export sizes.
Approval Board	Kanban columns: Draft, In Review, Needs Changes, Approved, Scheduled, Published.
Calendar	Month/week/list view, platform filter, campaign filter, thumbnails, media source labels, status badges.

Analytics	Post metrics, campaign comparison, best hook/format/time, AI recommendations.
Media Library	Upload image/video, generated asset history, thumbnails, source type, attach to post, delete/archive controls.
Reference Library	Upload reference designs, preview examples, tag style, view AI-extracted style notes, attach references to brand/campaign/post.

18. Task Division for Easier Development

If working alone, treat these as personal milestones. If working in a team, divide them by ownership.

Role / Workstream	Responsibilities	Recommended owner
Product + Research	User flow, feature priority, competitor analysis, documentation.	Sujit / Product lead
UI/UX Design	Wireframes, design system, dashboard, campaign flow, editor layout.	Designer or Sujit using Figma
Frontend Development	Next.js pages, components, forms, calendar UI, editor UI.	Frontend developer
Backend Development	APIs, database schema, auth, storage, scheduled jobs.	Backend developer
AI Integration	Prompt templates, AI planner, image generation, scoring, learning summary.	AI developer
Social API Integration	Meta app setup, page/IG connection, publishing, insights.	Integration developer
Testing + QA	User testing, API testing, failed posting retry, prompt output checks.	QA / all members
Documentation	Final report, architecture diagrams, setup guide, user manual.	Document owner
Media upload owner	Build file upload, storage, thumbnail preview, media metadata, and media-to-post linking.	Can be done by same developer after Product/Service profile module.
Reference design owner	Build reference upload, tagging, style analysis, and reference-to-campaign linking.	Can be done after Brand Kit and before advanced image generation.

19. Risks and Mitigation

Risk	Impact	Mitigation
Too broad scope	Project becomes unfinished.	Build image-first and Meta-first only in MVP.
AI output quality issues	Bad designs or text mistakes.	Manual editor + human approval + prompt templates.
Meta API approval complexity	Publishing may be delayed.	Start with test pages and simple posting; keep manual export fallback.
High AI cost	Product becomes expensive to run.	Limit generations per campaign, show credit usage, cache outputs.
Legal risk from copying Higgsfield	Brand/IP issues.	Use workflow inspiration only; create original UI, copy, and templates.
Competitor data overpromise	Loss of trust.	Only promise public creative observation and owned-account analytics.
User confusion with chat-only UX	Users do not know what to do.	Use guided forms, prompt controls, templates, and visual outputs.
Vendor dependency	Provider pricing/API changes hurt product.	Use model/provider abstraction from the beginning.
Uploaded media may not meet platform requirements	Scheduled post can fail if file type, size, duration, or account permission is invalid.	Validate media before scheduling, show platform-specific errors, and keep retry/failure status visible.
Reference designs may encourage copying	Legal/ethical risk if the system directly clones a competitor design, layout, logo, or campaign wording.	Use references only for style extraction; require user confirmation; generate original outputs and avoid protected logos/wording.

20. Success Metrics

Metric	What to measure	Why it matters
Time to first campaign	Minutes from signup to first approved campaign plan.	Measures onboarding quality.
Creative turnaround time	Time from brief to approved post.	Measures productivity improvement.
Approval rate	Generated variants approved vs rejected.	Measures output quality.
Manual edit frequency	Which fields users edit most.	Shows where AI/presets need

		improvement.
Posting reliability	Scheduled posts published successfully.	Measures trust in scheduler.
Engagement improvement	Performance of AI-assisted posts over time.	Measures product value.
Retention	Users returning weekly/monthly.	Measures business viability.
Uploaded media scheduling success	Percentage of user-uploaded image/video posts that publish successfully at scheduled time.	Confirms the platform is useful even when the user does not use AI generation.
Reference-guided generation satisfaction	Percentage of generated variants approved when a reference design is used.	Shows whether reference uploads improve creative alignment and reduce regeneration.

21. Product Roadmap

Phase	Focus	Features
Phase 1: MVP	Image-first creative workflow	Brand Kit, reference design upload, product profile, AI planner, image variants, user-uploaded media, manual editor, approval, calendar, Meta-first scheduling.
Phase 2: Learning loop	Performance feedback	Creative tags, Meta insights, best hook/style/time analysis, AI recommendations. Improve reference-based style memory and creative tagging.
Phase 3: Automation actions	Copilot workflows	Create 7-day plan, regenerate weak post, fill empty calendar, weekly report, competitor creative observation.
Phase 4: Video add-on	Premium creative expansion	Image-to-video, UGC-style video, provider abstraction, possibly Higgsfield/Runway/Kling.
Phase 5: Team/agency features	Collaboration and scale	Roles, comments, client approvals, white-label reports, billing, template marketplace.

22. Final Recommendation

The final build should be a visual marketing workspace with an AI copilot. The system should use structured campaign planning before generation, allow manual control after generation, require human approval before publishing, and connect performance data back to future creative decisions. This is more realistic and more defensible than building a pure Higgsfield clone.

- Build the product around the workflow: brand -> product -> campaign -> AI plan -> creative variants -> edit -> approve -> schedule -> publish -> learn.
- Keep the first version narrow: image-first and Meta-first.
- Use AI to reduce user effort, but use UI to maintain user control.
- Treat Higgsfield as a research benchmark and possible future video provider, not as the core moat.
- Make the moat the learning loop: approved brand memory + creative tags + performance data.
- Support both AI-generated and user-uploaded media so the system stays useful for businesses that already have photos, videos, or edited creatives.
- Support reference designs so users can show the system the visual direction they prefer instead of explaining everything through text prompts.
- Use uploaded references as style guidance only; the output should be original and brand-safe, not a direct copy of another design.

23. References and Research Sources

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